



Assignment 01

Name: Tayyaba Ziafat

Registration No: FA24B1-SE-018

Section: B

Course Name: Cloud Computing

Topic: Git, Gitea, GitHub, Git LFS, and GitHub Pages

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

Gitea is deployed as a lightweight self-hosted Git service using Docker Compose, running alongside a PostgreSQL database container (gitea_db). The curl command verifies that the local Gitea web service is actively responding and healthy by returning an HTTP status code of 200 on port 3000

```
tayyabaziafat@ubuntu:~/Gitea$ docker compose ps
NAME          IMAGE          COMMAND                  SERVICE    CREATED        STATUS        PORTS
gitea         gitea/gitea:latest "/usr/bin/entrypoint..." gitea      16 minutes ago Up 16 minutes 22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db      postgres:17-alpine "docker-entrypoint.s..." db         16 minutes ago Up 16 minutes 5432/tcp
tayyabaziafat@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
tayyabaziafat@ubuntu:~/Gitea$ _
```

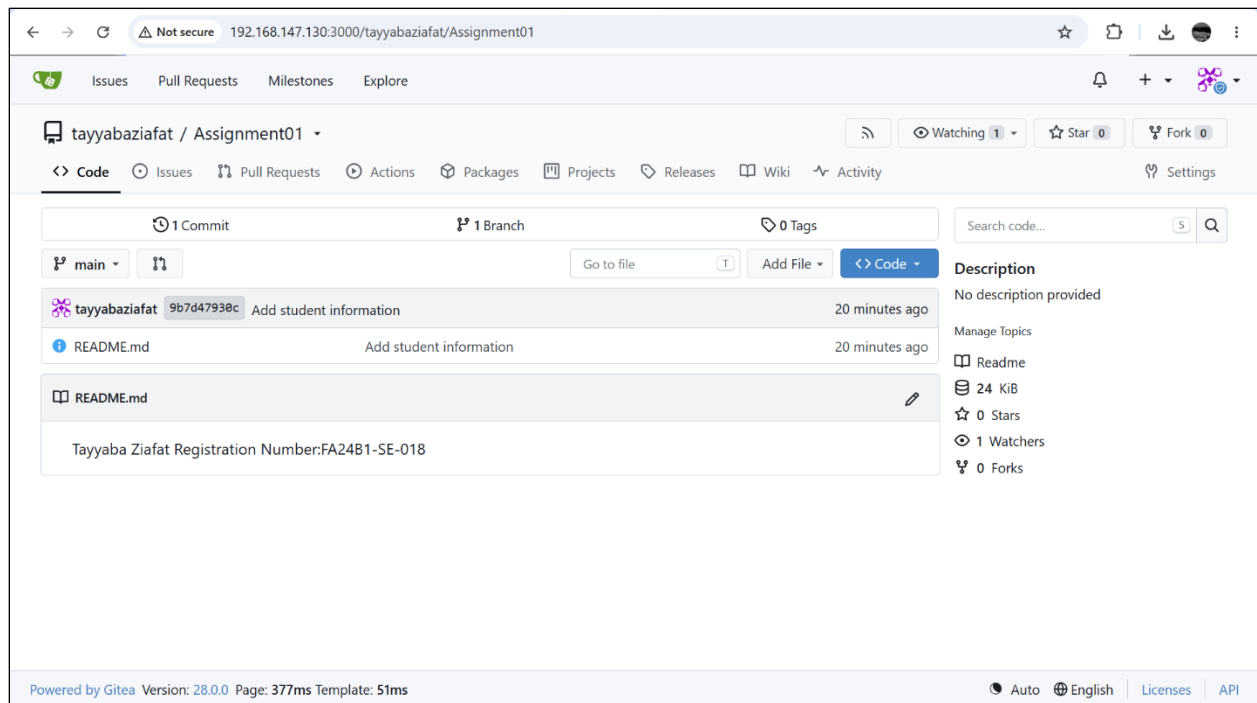
Step 1.2: Create and Push the Initial Repository

The local Git repository is pushed to the self-hosted Gitea remote server using the git push -u gitea main command. The -u (upstream) flag links the local main branch to the remote gitea repository, enabling streamlined future pulls and pushes.

```
tayyabaziafat@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://tayyabaziafat@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 277 bytes | 69.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/tayyabaziafat/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 1.3: Verify the Gitea Repository

Accessing the self-hosted Gitea web interface via IP address (192.168.147.130:3000) confirms that the repository structure, commit history, and README.md file containing student credentials have been successfully synchronized.



Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

The git remote -v command lists all configured remote endpoints for the local repository, displaying separate fetch and push URLs for both the local gitea instance and the cloud-based github repository.

```
tayyabaziafat@ubuntu:~/Assignment01$ git remote -v
gitea http://tayyabaziafat@127.0.0.1:3000/tayyabaziafat/Assignment01.git (fetch)
gitea http://tayyabaziafat@127.0.0.1:3000/tayyabaziafat/Assignment01.git (push)
github https://github.com/tayyabaziafat/Assignment01.git (fetch)
github https://github.com/tayyabaziafat/Assignment01.git (push)
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 2.2: Push to GitHub

Executing git push -u github main securely uploads the local commit history and branch structure to the remote GitHub repository over HTTPS, establishing GitHub as an additional upstream tracking target.

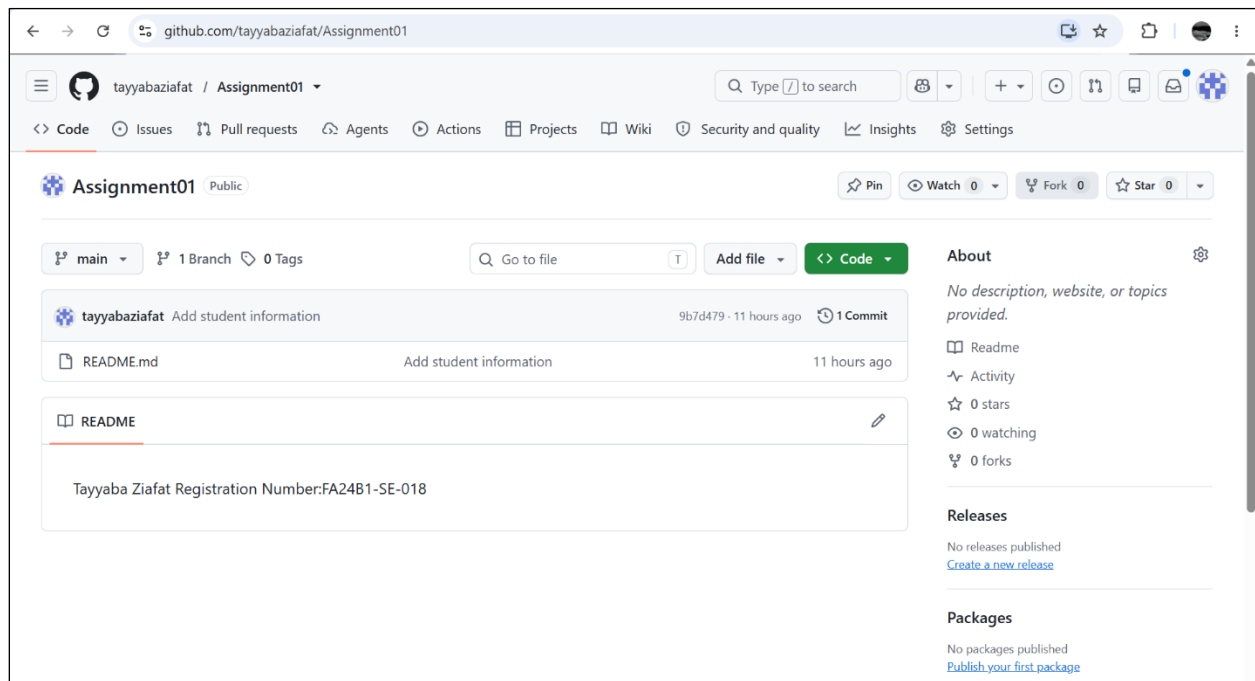
```
tayyabaziafat@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 277 bytes | 55.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com:tayyabaziafat/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Step 2.3: Verify the GitHub Repository

Browsing to

github.com/tayyabaziafat/Assignment01

verifies that the repository is publicly accessible and that all files and commit metadata match the local repository state.



Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

Git Large File Storage (LFS) replaces large binary files with text pointer references inside the Git history while storing the actual file content on a remote server. Running `git lfs track "*.bin"` updates the `.gitattributes` file to register all binary files under LFS management.

```
tayyabaziafat@ubuntu:~/Assignment01$ cd ~/Assignment01
tayyabaziafat@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
tayyabaziafat@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
tayyabaziafat@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
tayyabaziafat@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
tayyabaziafat@ubuntu:~/Assignment01$
```

Step 3.2: Create and Verify Three LFS Files

Dummy binary files exceeding 100MB are generated using the truncate command, staged into Git, and verified using git lfs ls-files --size to ensure they are properly tracked by LFS with their respective object hashes and file sizes.

```
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
tayyabaziafat@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
tayyabaziafat@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
tayyabaziafat@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
tayyabaziafat@ubuntu:~/Assignment01$
```

Step 3.3: Commit and Push the LFS Files

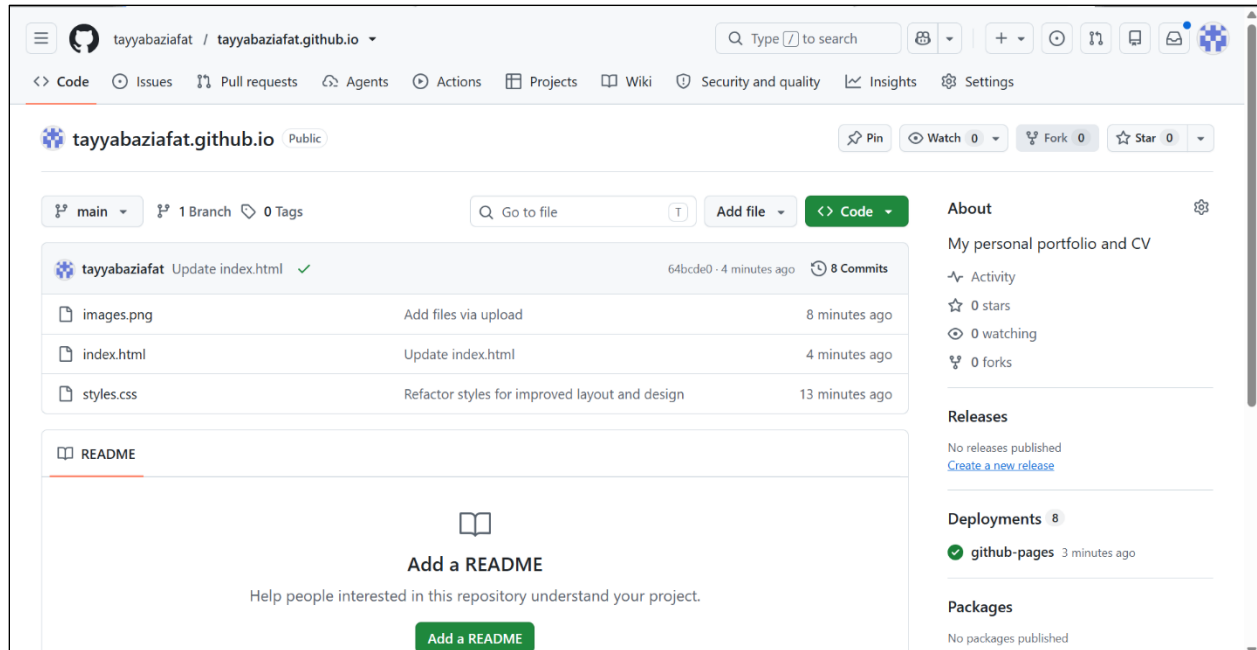
Pushing to GitHub via git push github main initiates the LFS client protocol, which uploads the large binary payloads to the LFS storage server before pushing the lightweight pointer commits to the standard Git repository.

```
tayyabaziafat@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Username for 'https://github.com': tayyabaziafat
Password for 'https://tayyabaziafat@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 2.2 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 802 bytes | 267.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com:tayyabaziafat/Assignment01.git
  9b7d479..41c2841 main -> main
tayyabaziafat@ubuntu:~/Assignment01$ _
```

Task 4: Create a Portfolio or CV with GitHub Pages

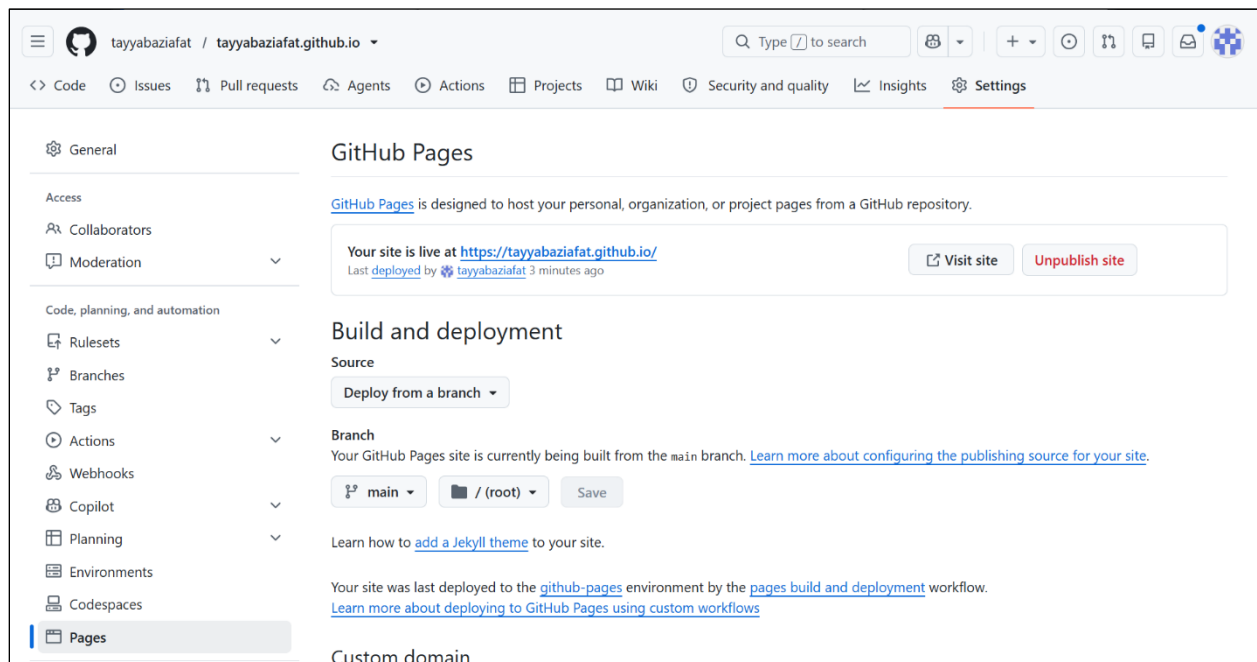
Step 4.1: Create and Push the Portfolio Repository

A specialized repository named tayyabaziafat.github.io is created to host a static personal website, containing core web assets including index.html, styles.css, and image files.



Step 4.2: Publish with GitHub Pages

GitHub Pages is configured under the repository settings to automatically build and deploy static web content directly from the root directory of the main branch.



Step 4.3: Verify the Live Portfolio

Navigating to tayyabaziafat.github.io confirms the successful deployment and rendering of the personal portfolio website, displaying professional profile details, education, technical skills, and project portfolios.

